A Subject-Specific Time Window Selection Method for Motor Imagery BCI

Yibing Zhang and Wenshan Zhao
School of Electronic and Information Engineering, Beijing Jiaotong University, Beijing, China
E-mail: Yibing Zhang (21120026@bjtu.edu.cn), Wenshan Zhao (wshzhao@bjtu.edu.cn)

2024 International Conference on Consumer Electronics - Taiwan (ICCE-Taiwan) | 979-8-3503-8684-4/24/$31.00 ©2024 IEEE | DOI: 10.1109/ICCE-Taiwan62264.2024.10674468

***Abstract*—Brain-computer interface based on motor imagery (MI) electroencephalogram is a promising technology for the future. However, the time latency during the MI period exhibits variability among the trials of different subjects, which can significantly affect the segmentation of each subject's trial using the fixed time window. To address this issue, the study proposes a subject-specific time window selection (TWS) method based on wavelet transform. The experiment results show that the proposed TWS achieves the highest accuracy across all three MI classification methods, with classification accuracies of 77.19%, 83.98%, and 85.33% on average respectively.**

***Keywords*—Brain-computer interface, motor imagery, wavelet transform, time-frequency image, time window selection**

I. Introduction

Brain-computer interface (BCI) is a system that enables the human brain to communicate directly with external computers or machines [1]. Currently, the research in BCI utilizing scalp electroencephalogram (EEG) has attracted much attention due to its non-invasion and low cost. Among EEG-based BCI systems, the motor imagery (MI)-BCI stands out as it solely relies on the subject's spontaneous brain potential, eliminating the need for external stimuli. This approach is considered a promising paradigm within BCI systems. When the subject performs the imagery task, the energy within a specific frequency band of the EEG signal undergoes regular changes. Specifically, a decrease in band energy is referred to as event-related desynchronization (ERD), while an increase in band energy is termed event-related synchronization (ERS) [2].

Classification of EEG for decoding motor intention is the key task in MI-BCI system. The majority of existing methods employ a fixed-length and fixed-position time window for the selection of EEG data during feature extraction and classification across all subjects [3]. However, the specific time at which different subjects begin to imagine varies, and the effective duration of the ERD/ERS phenomenon during MI also differs among subjects. Due to interference from data unrelated to MI task, aforementioned methods may result in poor classification results.

In this study, a subject-specific time window selection (TWS) method based on wavelet transform is proposed. First, multi-channel EEG signals are transformed into two-dimensional time-frequency images by wavelet transform. Second, the time-frequency images are used to determine the specific time window corresponding to MI task for each subject based on ERD phenomenon. Experiment results show that the proposed TWS method can achieve high MI classification performance by selecting the subject-specific time window.

This work was supported by Beijing Natural Science Foundation under Grant 4242010, and National Natural Science Foundation of China under Grant 61504008.

II. Materials and Methods

A. Dataset

To verify the effectiveness of TWS, experiments for MI classification have been conducted on the public dataset BCI Competition IV Dataset 2b [4]. The dataset 2b contains two classes of motor imagery samples of left-hand and right-hand movements form 9 subjects (B01 to B09). The data were recorded at the sampling frequency of 250 Hz with 3 electrodes (C3, Cz, and C4). There are five sessions for each subject, in which the first three sessions are training set, and the last two sessions are test set.

B. Proposed Method

Based on wavelet transform, this paper proposes a subject-specific TWS to address the problem of time window localization in MI-BCI system.

The ERD phenomenon of left-hand and right-hand motor imagery occurs in the corresponding contralateral sensorimotor cortex, i.e., the C4 and C3 regions of the cerebral cortex [5]. During left-hand motor imagery, the energy in the corresponding C4 region on the right side of the brain decreases. Meanwhile, during right-hand motor imagery, the energy in the corresponding C3 region on the left side of the brain decreases. In this context, the ERD phenomenon is leveraged to select the time window of EEG signal related to certain MI task customized for each subject.

Wavelet transform is a widely used time-frequency analysis method for non-stationary signal processing. In this study, the continuous wavelet transform (CWT) is employed to extract the time-frequency representation of EEG data to capture the ERD phenomenon related to certain MI task. The CWT has the capability of multi-scale analysis and circumvents the limitations associated with the fixed time window of the short-time Fourier transform, which can be expressed by [6]

$$CWT(a,\tau)=\frac{1}{\sqrt{a}}\int_{-\infty}^{\infty} f(t)\psi(\frac{t-\tau}{a})dt \tag{1}$$

where $f(t)$ is the input signal, ψ denotes the wavelet function, a and τ represent the scale factor and the position factor, respectively.

Generally, the proposed TWS method mainly includes three steps.

(1) The EEG data from the C4 channel are averaged for all left-hand MI tasks performed by a subject. Analogously, the EEG data from the C3 channel are averaged for all right-hand MI tasks undertaken by the same subject.

(2) Select the appropriate wavelet function for CWT to obtain the time-frequency images of MI EEG.

(3) Select the subject-specific time window determined by the time period of energy decrease on the time-frequency image based on ERD phenomenon.

III. Experimental Results

Taking the subject B01 as an example, the time-frequency image of wavelet transform is shown in Fig. 1. It can be seen that there is a sudden decrease in energy within the time range of 3.2s to 7.0s. Therefore, this study regards this time period as the specific time window of the subject B01.

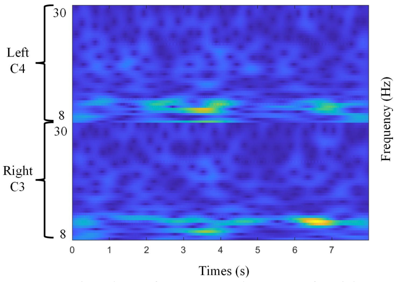

Fig. 1. Wavelet time-frequency image of subject B01.

This study selected four different wavelet functions to analyze the EEG signal of the subject B01, namely Complex Morlet (cmor), Symlets, Coiflets and Morlet. The comparison results are shown in Fig. 2. It can be seen that the energy of the time-frequency image based on cmor4-4 is the most concentrated. Therefore, cmor4-4 is selected for subsequent experiments in this study.

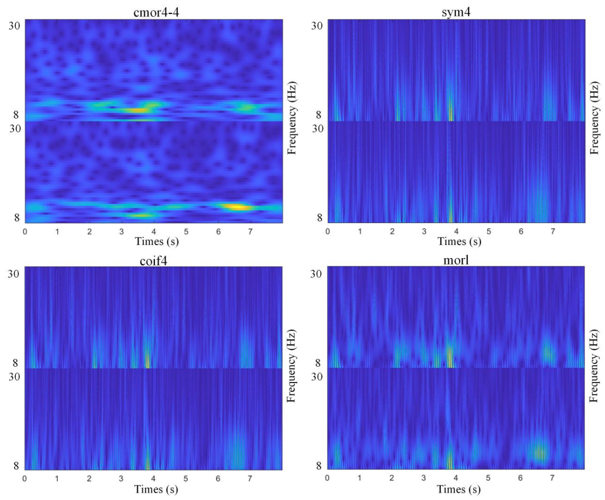

Fig. 2. Time-frequency images using different wavelet functions.

Experiments have been conducted on three representative MI classification methods to prove the superiority of the TWS method proposed in this study. These methods include common spatial pattern (CSP) [7], EEGNet [8], and Conformer [9]. The programming language used in this study is Python. The loss function is cross-entropy. The Adam optimizer is utilized to train the classification model for 300 epochs. The learning rate is 0.0001 and the batch size is 256. The classification accuracy is used as the metric for performance evaluation.

Table I shows the performance comparison of subject-specific time window with fixed time window when using different MI classification methods. The best performance of each method is highlighted in boldface. It can be seen that the proposed method can achieve the best performance across all subjects.

IV. Conclusion

This paper proposes a subject-specific time window selection method for MI classification. Firstly, wavelet transform is performed on EEG signals to obtain the time-frequency images. Secondly, the optimal time window is selected based on ERD phenomenon. Experiment results demonstrate that the selection of optimal time window customized for each subject significantly enhances the classification accuracy in MI-BCI.

References

[1] G. Schalk, D. J. McFarland, T. Hinterberger, N. Birbaumer, and J. R. Wolpaw, “BCI2000: A general-purpose brain-computer interface (BCI) system,” IEEE Trans. Biomed. Eng., vol. 51, no. 6, pp. 1034–1043, Jun. 2004.

[2] G. Pfurtscheller and F. H. L. da Silva, “Event-related EEG/MEG synchronization and desynchronization: basic principles,” Clin. Neurophysiol., vol. 110, no. 11, pp. 1842–1857, Nov. 1999.

[3] X. Deng, B. Zhang, N. Yu, K. Liu, and K. Sun, “Advanced TSGL-EEGNet for Motor Imagery EEG-Based Brain-Computer Interfaces,” IEEE Access, vol. 9, pp. 25118–25130, 2021.

[4] R. Leeb, C. Brunner, and G. Müller-Putz. BCI Competition 2008—Graz Data Set B. Accessed: Jun. 25, 2019. [Online]. Available: http://www.bbci.de/competition/iv/

[5] G. Pfurtscheller, C. Brunner, A. Schlögl, and F. H. L. da Silva, “Mu rhythm (de)synchronization and EEG single-trial classification of different motor imagery tasks,” Neuroimage, vol. 31, no. 1, pp. 153–159, May 2006.

[6] M. Wacker and H. Witte, “Time-frequency techniques in biomedical signal analysis a tutorial review of similarities and differences,” Methods Inform. Med., vol. 52, pp. 279–296, 2013.

[7] H. Higashi and T. Tanaka, “Simultaneous design of FIR filter banks and spatial patterns for EEG signal classification,” IEEE Trans. Biomed. Eng., vol. 60, no. 4, pp. 1100–1110, Apr. 2013.

[8] V. J. Lawhern, A. J. Solon, N. R. Waytowich, S. M. Gordon, C. P. Hung, and B. J. Lance, “EEGNet: a compact convolutional neural network for EEG-based brain-computer interfaces,” J. Neural Eng., vol. 15, no. 5, Oct. 2018, no. 056013.

[9] Y. Song, Q. Zheng, B. Liu, and X. Gao, “EEG Conformer: Convolutional Transformer for EEG Decoding and Visualization,” IEEE Trans. Neural Syst. Rehabil. Eng., vol. 31, pp. 710–719, 2023.

TABLE I. Classification Accuracies (%) on BCI Competition IV Dataset 2B

Subject	**CSP**			**EEGNet**			**Conformer**		
	3s-7s	***4s-8s***	***TWS***	***3s-7s***	***4s-8s***	***TWS***	***3s-7s***	***4s-8s***	***TWS***
B01(3.2s-7.0s)	65.00	64.38	**65.94**	72.88	68.63	**77.38**	76.39	80.56	**81.25**
B02(3.6s-7.1s)	58.21	55.71	**60.36**	64.43	59.34	**66.58**	64.71	60.30	**67.65**
B03(3.0s-7.2s)	52.50	52.50	**53.44**	85.50	79.88	**85.62**	81.95	79.17	**84.73**
B04(3.5s-7.0s)	90.31	87.19	**93.44**	90.50	86.13	**91.75**	93.25	91.92	**94.60**
B05(3.4s-6.6s)	73.75	77.19	**77.81**	84.25	79.88	**84.75**	85.68	81.09	**87.03**
B06(3.4s-7.4s)	83.75	85.94	**86.56**	85.81	80.15	**86.13**	83.62	82.23	**86.39**
B07(3.8s-7.0s)	75.94	74.69	**78.75**	74.56	76.13	**81.75**	79.17	80.56	**83.95**
B08(3.2s-6.4s)	89.38	82.81	**90.31**	92.13	91.50	**93.38**	89.64	90.95	**92.27**
B09(3.5s-7.2s)	80.94	87.50	**88.13**	86.25	87.19	**88.50**	86.39	87.56	**90.06**
Average	74.42	74.21	**77.19**	81.81	78.76	**83.98**	82.31	81.59	**85.33**